RUTHVIJ RANE

Parel, Mumbai • 9619509972 • ruthvijrane2001@gmail.com
LinkedIn: ruthvij-rane-b758111b4/ GitHub: github.com/ruthrane25
Portfolio: https://ruthportfoliofive.vercel.app/

SUMMARY

Software Engineer with strong expertise in Python, desktop application development, backend systems, and data processing. Proven experience building real-time control software, custom data parsers (G-code/DXF), and full-stack web applications using Python (PyQt/PySide), JavaScript, React, Node.js, and MongoDB. Skilled in hardware-software integration, system automation, and transforming complex data workflows into intuitive user interfaces. Strong problem-solving background with a practical approach to designing, developing, testing, and optimizing scalable software solutions.

TECHNICAL SKILLS

Languages : Python, Java, JavaScript, TypeScript, SQL, HTML, CSS

Data & Analytics / File Processing : Data Processing, Pandas, NumPy, CSV/JSON Processing, Data Visualization, G-code Parsing, DXF Parsing

Software & Desktop Development : Python, PySide6, PyQt, Qt, Desktop Application Development, GUI Development, PyInstaller, Inno Setup

Web & Backend Development : React.js, Node.js, Express.js, SvelteKit, Tailwind CSS, RESTful APIs, API Integration

Databases : MySQL, MongoDB, Database Integration

Systems & Automation : Serial/USB Communication, G-code, Hardware-Software Integration, Automation, Debugging & Troubleshooting

Tools & Practices : Git, GitHub, SVN, Qt Designer, Software Deployment, Agile Methodologies

WORK EXPERIENCE

Software Design Engineer , Specialize Products Pvt. Ltd. **May 2025 - Present**
Mumbai,India

- Architected & Developed Desktop Control Software: Engineered full-featured desktop applications and responsive graphical user interfaces using Python (PyQt/PySide) for industrial laser systems, photovoltaic scribers, and 3D bio-printers, ensuring intuitive real-time monitoring and instrument control.
- Built Data Parsing & Processing Engines: Developed custom data processing modules and parsers (G-code, DXF) to transform complex vector files and structural designs into high-precision machine execution paths and spatial coordinate arrays.
- Hardware-Software Integration & Communication: Implemented robust serial and USB communication pipelines for real-time command transmission, multi-axis motion control, and live operational telemetry feedback from machine controllers.
- Automated Thermal & Inspection Software: Designed software logic for automated laser exposure systems and synchronized non-contact thermal mapping modules, streamlining data acquisition and analysis workflows.

Web Developer Intern , VC Technosolutions Pvt. Ltd. **April 2024 - April 2025**
Navi Mumbai,India

- Full-Stack Automation Lead: Directed a 3-engineer team to build a centralized operations platform using SvelteKit, TypeScript, and MongoDB, replacing manual Excel tracking and boosting efficiency by 25%.
- Frontend Architecture & UI: Developed a high-performance, responsive interface with Tailwind CSS, streamlining internal workflows and user engagement.
- Real-Time Analytics & Reporting: Designed real-time tracking modules and automated reporting pipelines, enhancing operational visibility and productivity.

Logistics Team Leader, Flipkart (Ekart) — **May - Dec 2023**, Mumbai,India

- Managed end-to-end inventory efficiency and financial control.
- Minimized inventory losses and maintained variance within budget.
- Ensured timely and accurate replenishment and stock-taking to prevent productivity loss.

EDUCATION

B.E. - Electronics Engineering | 7.15 CGPA — **2018 - 2022**, Navi Mumbai,India
University of Mumbai, Pillai College of Engineering

HSC | 65 % — **2017 - 2018**, Mumbai,India
Maharishi Dayanand College of Science, Parel

SSC | 78 % — **2015 - 2016**, Mumbai,India
Holy Cross High School, Lower Parel

PROJECTS

Industrial Laser Scribing & Material Processing Software | Python, PyQt, Custom Data Parsing, G-code/DXF

- Engineered a cross-platform desktop control suite using Python (PyQt/PySide) to operate high-precision thin-film photovoltaic (solar) laser scribers and semiconductor wafer patterners.
- Developed custom data parsing engines to transform complex G-code and DXF vector files into precise spatial toolpaths and real-time execution coordinates.
- Implemented asynchronous serial and USB communication layers for live machine telemetry, multi-axis motion control, and operational feedback.

VC Tech Operations Platform | SvelteKit, TypeScript, Node.js, MongoDB, Tailwind CSS

- Directed a 3-engineer team to architect and deploy a centralized internal operations platform, automating legacy Excel-based tracking and boosting operational efficiency by 25%.
- Developed a high-performance, responsive UI with SvelteKit and Tailwind CSS backed by real-time analytics dashboards and automated reporting pipelines.

Drone Delivery System (Final Year Project)| Python, Autonomous Navigation, System Integration, [Link]

- Designed and developed an autonomous drone delivery prototype featuring real-time obstacle and collision avoidance algorithms.
- Assessed the feasibility and operational logistics of integrating delivery drones into automated supply chains, highlighting high-utility use cases during quarantine and emergency scenarios.

VOLUNTEERING & LEADERSHIP

IEEE (Institute of Electrical and Electronics Engineers) PCE — **2019 - 2022**, Navi Mumbai,India

- Social Media Head (2021 - 2022).
- Graphics Joint-Head (2020 - 2021)

LANGUAGES

- English (Professional Working Proficiency)
- Hindi (Full Professional Proficiency)
- Marathi (Native or Bilingual Proficiency)